

Complementation and clause size in Seediq

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roadmap

1. introduction

2. theoretical assumptions

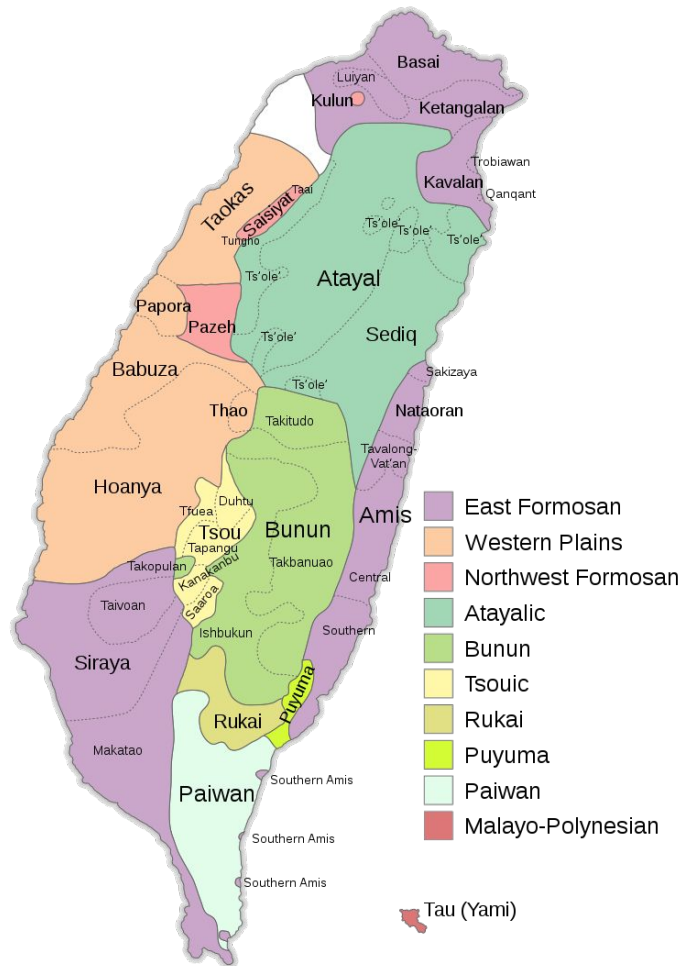
3. empirical data

- Austronesian voice systems
- complementation

4. discussion

- implicational scale
- functional decomposition

5. conclusion





main takeaway

the **Austronesian voice system** is a complex phenomenon that should be decomposed across multiple functional projections

theoretical framework:

Implicational Complementation Hierarchy (ICH):

there is an implicational scale in linguistic properties across different clause types, corresponding to syntactic structure

(Wurmbrand & Lohninger 2023)

this talk:

four tiers of complement clauses for verbal morphology in Seediq...

1) gerunds	vP	bare verb
2) control clauses	?P	AV marking, no TAM
3) tough-constructions	?P	AV/PV/LV/CV, no TAM
4) finite clauses	CP	AV/PV/LV/CV, TAM

Austronesian voice systems

- **voice morphology**: indicates which argument—core or non-core—is the pivot
- **pivot**: privileged argument w.r.t. definiteness, word order, case, extraction, etc.

(1) ‘The man hunts wild pigs in the mountains with a spear.’

- | | | |
|----|---|---------------------------|
| a. | q⟨ m ⟩alup [a caucau] tua vavuy i gadu tua vulu | actor voice (AV) |
| | ⟨AV⟩hunt PIV man OBL pig LOC mountain OBL spear | |
| b. | qalup- en nua caucau [a vavuy] i gadu tua vulu | patient voice (PV) |
| | hunt-PV GEN man PIV pig LOC mountain OBL spear | |
| c. | qalup- an nua caucau tua vavuy [a gadu] tua vulu | locative voice (LV) |
| | hunt-LV GEN man OBL pig PIV mountain OBL spear | |
| d. | si -qalup nua caucau tua vavuy i gadu [a vulu] | circumstantial voice (CV) |
| | CV-hunt GEN man OBL pig LOC mountain PIV spear | |

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theoretical assumptions

complements have different **degrees of independence** based on morphological, syntactic, and semantic properties

“**Binding Hierarchy**”

(Givon 1980)

“**Implicational Complementation Hierarchy**”

(Wurmbrand & Lohninger 2023)

MOST CLAUSAL <---

Class 1 >> Class 2 >> Class 3

--> LEAST CLAUSAL

properties of complement clauses:

- tense/aspect/mood marking on embedded verb
- overt subject and case marking in complement clause
- overt complementizer and adverbial modification
- A vs. A' movement out of complement clause



Implicational Complementation Hierarchy (ICH)

there are three main classes of complements:

Proposition >> Situation >> Event

operator domain (CP) >> TAM domain (TP) >> theta domain (VP)

signature effect:

there is an **implicational scale** along the hierarchy:

- in terms of integration, transparency, complexity, etc.
- the hierarchy is ***universal***, but the properties are ***relative***
- if a class of complements gains/loses a property, any class of complements further along the scale will have the same value
- some verb can have complements of multiple classes

	<i>Proposition</i>	<i>Situation</i>	<i>Event</i>
overt subject	✓	✓	*
infinitival complement	*	✓	✓
clitic climbing	*	*	✓

Table 1: linguistic properties in Serbian complement clauses

	<i>Proposition</i>	<i>Situation</i>	<i>Event</i>
Brazilian Portuguese	*	*	*
Italian	*	*	✓
Czech	*	✓	✓

Table 2: clitic climbing out of complement clauses

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Seediq complements

from most to least independent complement clauses...

finite >> **tough** >> **control** >> **gerund**

- | | | |
|-------------------------------|-----------|--|
| 1) gerunds | vP | e.g. 'ask', 'begin', 'try', etc. |
| 2) control clauses | ?P | e.g. 'ask', 'begin', 'try', 'forget', 'decide', etc. |
| 3) tough-constructions | ?P | e.g. 'easy', 'difficult', 'good', 'hard', etc. |
| 4) finite clauses | CP | e.g. 'say', 'know', 'believe', etc. |

properties that were tested for...

on embedded verb: **voice morphology, TAM morphology**

embedded clause: **overt subject, long object movement**



	<i>finite</i>	<i>tough</i>	<i>control</i>	<i>gerund</i>
i. AV marking	✓	✓	✓	*
ii. non-AV marking	✓	✓	*	*
iii. TAM marking	✓	*	*	*
iv. overt subject	✓	*	*	*
v. object movement	✓	✓	✓	*

Table 1: properties of embedded clauses in Seediq



	finite	tough	control	gerund
i. AV marking	✓	✓	✓	*
ii. non-AV marking	✓	✓	*	*
iii. TAM marking	✓	*	*	*
iv. overt subject	✓	*	*	*
v. object movement	✓	✓	✓	*



embedded verb is...

gerund: bare

control: AV only
no TAM

tough: AV/PV/LV/CV
no TAM

finite: AV/PV/LV/CV
yes TAM

Table 1: properties of embedded clauses in Seediq

embedded verb is...

gerund: bare verb

control: AV only, no TAM

- (2) a. prading ka [gerund **phuma**=nami paye npalic] *⟨m⟩huma/phma-un
begin.AV PIV plant=1PL.GEN rice glutinous ⟨AV⟩plant/plant-PV
'[Our planting of rice] begins.'
- b. d⟨m⟩urun [gerund **kbarux**=su pila] ka Bakan *k⟨m⟩⟨n⟩barux
⟨AV⟩ask lend=2SG.GEN money PIV Bakan ⟨PFV⟩⟨AV⟩lend
'Bakan is asking for [your lending of money].'
- (3) a. prading=nami [control ⟨m⟩**huma** paye npalic] *phma-un
begin.AV=1PL.PIV ⟨AV⟩plant rice glutinous plant-PV
'We will begin [to plant rice].'
- b. d⟨m⟩urun isu [control **k⟨m⟩barux** pila] ka Bakan *k⟨m⟩⟨n⟩barux
⟨AV⟩ask 2SG.OBL ⟨AV⟩lend money PIV Bakan ⟨PFV⟩⟨AV⟩lend
'Bakan is asking you [to lend money].'

embedded verb is...

tough: AV/PV/LV/CV, no TAM

finite: AV/PV/LV/CV, yes TAM

- (4) a. rriqu=mu [tough **slhya-an**] ka patis nii *s⟨n⟩lhya-an
difficult.AV=1SG.GEN read-LV PIV book this ⟨PFV⟩read-LV
'The book is difficult for me [to read].'
- b. ⟨n⟩malu [tough **puq-un**] ka ido nii *p⟨n⟩uq-un
⟨PFV⟩good.AV eat-PV PIV rice this ⟨PFV⟩eat-PV
'The food was good [to eat].'
- (5) a. ⟨m⟩kela=ku [finite ⟨m⟩⟨n⟩**osa** Plsiya ka Awi]
⟨AV⟩know=1SG.PIV ⟨PFV⟩⟨AV⟩go Puli PIV Awi
'I know [that Awi went to Puli].'
- b. ⟨m⟩kela=ku [finite ⟨n⟩**sa-an** Plsiya ka Awi]
⟨AV⟩know=1SG.PIV go-LV Puli PIV Awi
'I know [that Awi went to Puli].'



	<i>finite</i>	<i>tough</i>	<i>control</i>	<i>gerund</i>
i. AV marking	✓	✓	✓	*
ii. non-AV marking	✓	✓	*	*
iii. TAM marking	✓	*	*	*
iv. overt subject	✓	*	*	*
v. object movement	✓	✓	✓	*

embedded clause has...

gerund: POSS subject

control: PRO subject

tough: PRO subject (?)

finite: overt subject

Table 1: properties of embedded clauses in Seediq

embedded clause has...

gerund: POSS subject

control: PRO subject

- (2) a. prading ka [gerund phuma=**nami** paye npalic]
begin.AV PIV plant=1PL.GEN rice glutinous
'[Our planting of rice] begins.'
- b. d⟨m⟩urun [gerund kbarux=**su** pila] ka Bakan
⟨AV⟩ask lend=2SG.GEN money PIV Bakan
'Bakan is asking for [your lending of money].'
- (3) a. prading=**nami**; [control ⟨m⟩huma **PRO**; paye npalic]
begin.AV=1PL.PIV ⟨AV⟩plant rice glutinous
'We will begin [to plant rice].'
- b. d⟨m⟩urun **isu**; [control k⟨m⟩barux **PRO**; pila] ka Bakan
⟨AV⟩ask 2SG.OBL ⟨AV⟩lend money PIV Bakan
'Bakan is asking you [to lend money].'

embedded clause has...

tough: PRO subject (?)

finite: overt subject

(4) a. rriq=mu; [tough slhya-an PRO_i] ka patis nii
difficult.AV=1SG.GEN read-LV PIV book this
'The book is difficult for me [to read].'

b. ⟨n⟩malu [tough puq-un PRO_i] ka ido nii
⟨PFV⟩good.AV eat-PV PIV rice this
'The food was good [to eat].'

(5) a. ⟨m⟩kela=ku [finite ⟨m⟩⟨n⟩osa Plsiya ka Awi]
⟨AV⟩know=1SG.PIV ⟨PFV⟩⟨AV⟩go Puli PIV Awi
'I know [that Awi went to Puli].'

b. ⟨m⟩kela=ku [finite ⟨n⟩sa-an Plsiya ka Awi]
⟨AV⟩know=1SG.PIV go-LV Puli PIV Awi
'I know [that Awi went to Puli].'



	<i>finite</i>	<i>tough</i>	<i>control</i>	<i>gerund</i>
i. AV marking	✓	✓	✓	*
ii. non-AV marking	✓	✓	*	*
iii. TAM marking	✓	*	*	*
iv. overt subject	✓	*	*	*
v. object movement	✓	✓	✓	*



embedded clause is...

gerund: DP phase

control: transparent

tough: transparent (?)

finite: CP phase

(can be unlocked, R&R 2005)

Table 1: properties of embedded clauses in Seediq

embedded clause is...

gerund: DP phase → no long object movement

- (2) a. prading ka [_{gerund} phuma=nami **paye npalic**]
begin.AV PIV plant=1PL.GEN rice glutinous
'[Our planting of rice] begins.'
- *prding-un [_{gerund} phuma=nami] **ka paye npalic**
begin-PV plant=1PL.GEN PIV rice glutinous
- b. d⟨m⟩urun [_{gerund} kbarux=su **pila**] ka Bakan
⟨AV⟩ask lend=2SG.GEN money PIV Bakan
'Bakan is asking for [your lending of money].'
- *Bakan s-durun [_{gerund} kbarux=su] **ka pila**
Bakan cv-ask lend=2SG.GEN PIV money

embedded clause is...

control: transparent → yes long object movement

- (3) a. prading=nami [control ⟨m⟩huma **paye npalic**]
begin.AV=1PL.PIV ⟨AV⟩plante rice glutinous
'We will begin [to plant rice].'
prding-un=nami [control ⟨m⟩huma] **ka paye npalic** di
begin-PV=1PL.GEN ⟨AV⟩plant PIV rice glutinous ASP
- b. d⟨m⟩urun isu [control k⟨m⟩barux **pila**] ka Bakan
⟨AV⟩ask 2SG.OBL ⟨AV⟩lend money PIV Bakan
'Bakan is asking you [to lend money].'
Bakan s-durun isu [control k⟨m⟩barux] **ka pila**
Bakan CV-ask 2SG.OBL ⟨AV⟩lend PIV money

embedded clause is...

tough: transparent (?) → yes long object movement

- (4) a. ?rriqu(=ku) [*tough* ⟨m⟩slluhe patis nii]
difficult.AV=1SG.PIV ⟨AV⟩read book this
'The book is difficult for me [to read].'
rriqu=mu [*tough* slhay-an] ka patis nii
difficult.AV=1SG.GEN read-LV PIV book this
- b. ?⟨n⟩malu [*tough* ⟨m⟩ekan ido nii]
⟨PFV⟩good.AV ⟨AV⟩eat rice this
'The food was good [to eat].'
⟨n⟩malu [*tough* puq-un] ka ido nii
⟨PFV⟩good.AV eat-PV PIV rice this

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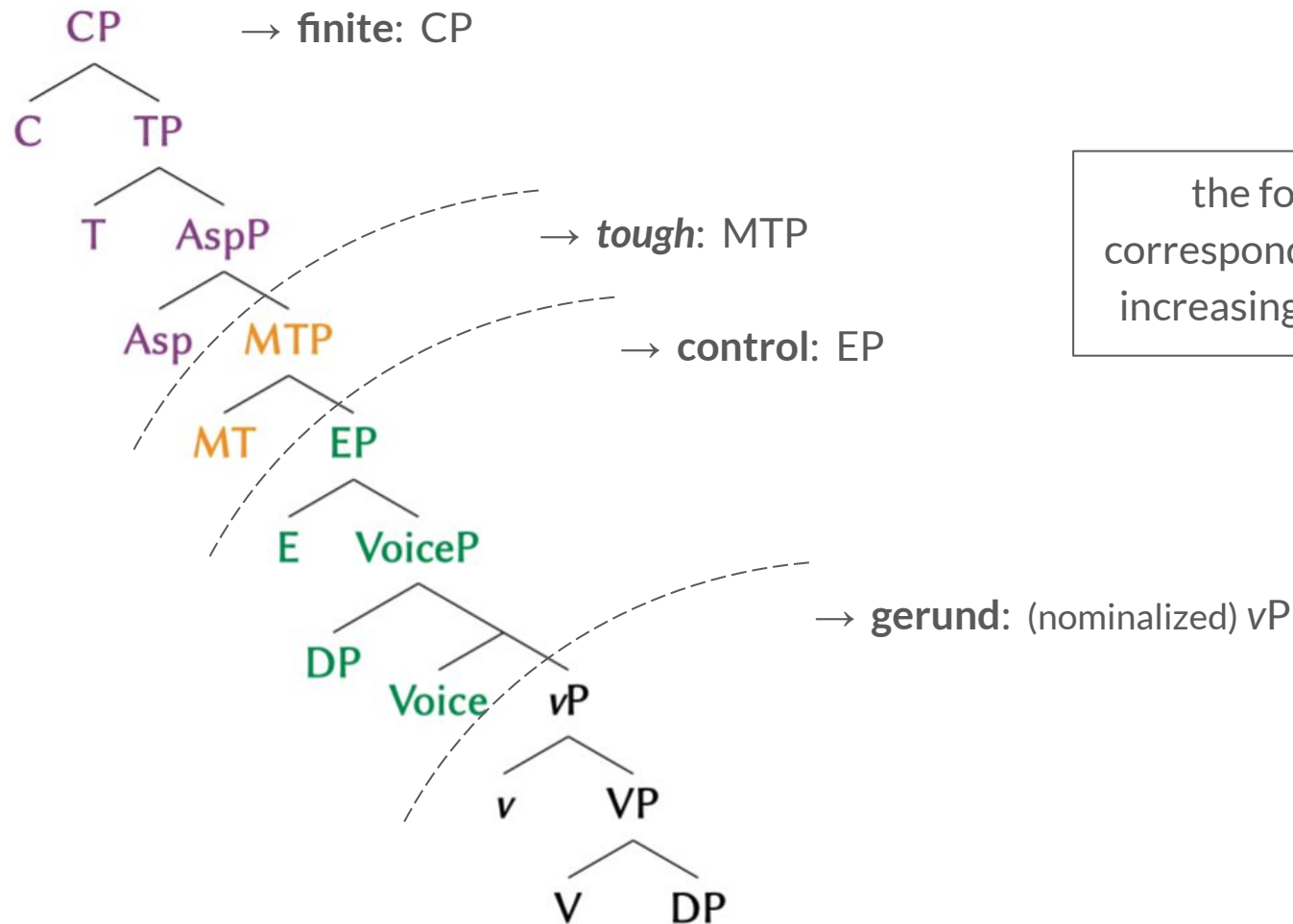


implicational hierarchy

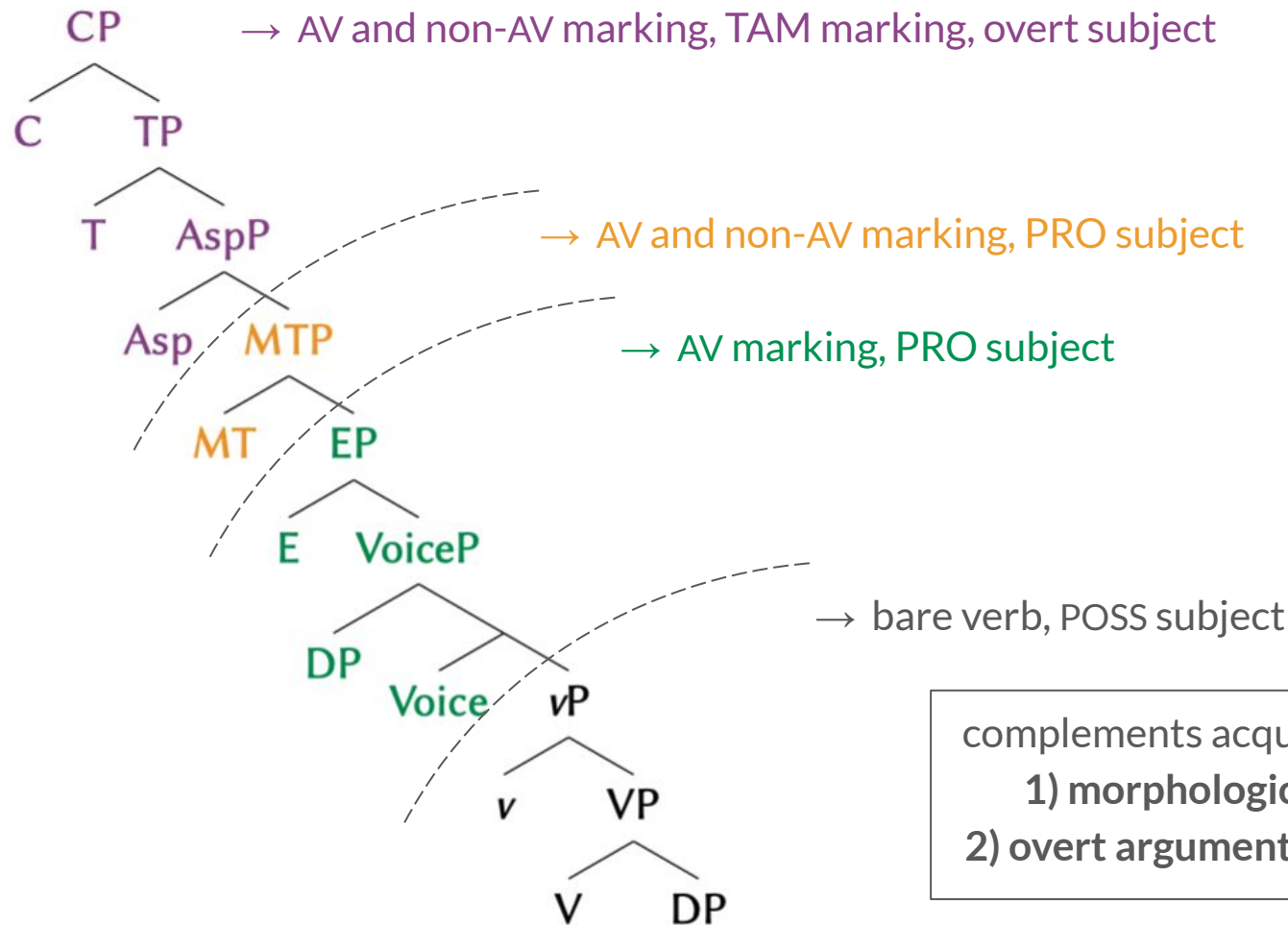
there are **four types of complement clauses** in Seediq, which form an implicational scale in terms of complexity and independence

	finite	tough	control	gerund
↑ <i>independence</i>				
↑ <i>complexity</i>				
↓ <i>transparency</i>				
↓ <i>integration</i>				
i. AV marking	✓	✓	✓	*
ii. non-AV marking	✓	✓	*	*
iii. TAM marking	✓	*	*	*
iv. overt subject	✓	*	*	*
v. object movement	✓	✓	✓	*

Table 1: properties of embedded clauses in Seediq



the four clause types
correspond to complements of
increasing **size** and **structure**



complements acquire properties such as

- 1) **morphological complexity** and
- 2) **overt arguments** with increase in size



functional projections

previous work on Austronesian voice support these projections...

M(iddle)T(opic)

a low topic-like projection—movement to middle field:

pivots have topic-like qualities; pivots move to edge of Voice domain

(Aldridge 2004; Pearson 2005; Rackowski & Richards 2005; V. Chen 2017; Y. Wu 2024)

E(vent)

an event closure projection—used in non-finite marking:

default “AV” marking in reduced clauses; non-finite *-a marker in PAn

(Starosta et al. 1982; T.C. Chen 2010; Travis 2010; Wurmbrand 2016; Aldridge 2021)



implications

C > T > Asp > MT > E > Voice > v > √ROOT

voice morphology:

- 1) AV is not a “morphological default (Chen 2010, Wurmbrand 2016)” form if bare verbs exists, and instead **reflects additional structure**
- 2) and non-AV reflects further additional structure on top of that
- 3) two levels of **defectiveness**: bare verbs and AV-marked verbs

clause structure:

- 4) AV/PV/LV/CV is located below TAM in the functional domain
- 5) **voice is located in the middle field**—*not* the left periphery (Chen 2017)—and reflects information structure and/or finiteness

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conclusion

the **Austronesian voice system** is a complex phenomenon that should be decomposed across multiple functional projections

- despite voice morphology being in complementary distribution
- evidence for further decomposition: **MT**, **E**, and **Appl** (Wu 2025)

functional hierarchy:

C > **T** > **Asp** > **MT** > **E** > **Voice** > **Appl** > **v** > $\sqrt{\text{ROOT}}$

four tiers of clauses for verbal morphology in Seediq...

1) gerunds	vP	bare verb
2) control clauses	EP	AV marking, no TAM
3) tough-constructions	MTP	AV/PV/LV/CV, no TAM
4) finite clauses	CP	AV/PV/LV/CV, TAM



why decompose?

other motivations
to decomposing voice...

- **different morphological orientation**—prefix vs. suffix:
morpheme-specific or structural difference?
- voices have **different syntactic distributions**,
cf. complementation, restructuring, adverbs

(V. Chen 2010, H. Chang 2017, Wurmbrand 2016, a.o.)

- **distinct historical origins** of voice markers:
*-en/-an as prepositions, *si- as light verb, etc.

(Starosta et al. 1981, Peterson 2017)

- imperative paradigm is **morphologically complex**

(Aldridge 2021, Wu 2025)

thank you!

special thanks to...

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